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2016-2017

M1 MACRO

Group A

Session 3

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Gen Eq. with a market for savings

$$\max \log c_1 + \beta \log c_2$$

$$BC_1 \quad c_1 + S = Y_1$$

choose
optimally
 c_1, c_2, S

$$BC_2 \quad c_2 = Y_2 + \underbrace{(1+r)}_{\text{price to be determined in gen. eq.}} S$$

$$BC_1 : S = Y_1 - C_1$$

$$BC_2 : C_2 = Y_2 + (1+r)(Y_1 - C_1)$$

$\Rightarrow$

Intertemporal Budget Constraint (IBC)

$$(1+r)C_1 + C_2 = \underbrace{(1+r)Y_1 + Y_2}_R$$

$\Rightarrow$

IBC

$$C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}$$

Sum of disc.
expenditures

Sum of discounted
endowments

$\Rightarrow$ New ph

$$\text{Max } \log C_1 + \beta \log C_2$$

sl.

$$C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}$$

Take the previous compl. eq.

$$\begin{cases} p_1 = 1 \\ p_2 = \beta^{y_2/y_1} \end{cases}$$

if $\frac{1}{1+r} = p_2$,

then r is the equilibrium interest rate.

$$r = \frac{1}{p_2} - 1 = \frac{y_2}{\beta y_1} - 1$$

Optimality condition

$$\frac{\beta/c_2}{1/c_1} = \frac{P_2}{P_1}$$

$$\Leftrightarrow \frac{1}{c_1} = \frac{P_1}{P_2} \beta \frac{1}{c_2}$$

$$\frac{1}{c_1} = (1+\alpha) \frac{1}{c_2}$$

Intertemp
Euler
Equilibrium

Introducing uncertainty

[optimal behavior of an agent
given prices]

Assume y_2 is uncertain
 $y_2 = \begin{cases} E(y_2) - 1 & \text{with prob } 1/2 \\ E(y_2) + 1 & \text{with prob } 1/2 \end{cases}$

check

$$\begin{aligned} E(y_2) &= \frac{1}{2} [E(y_2) + 1] + \frac{1}{2} [E(y_2) - 1] \\ &= 2 \cdot \frac{1}{2} E(y_2) + \frac{1}{2} - \frac{1}{2} \\ &= E(y_2) \end{aligned}$$

It makes sense to assume that
with uncertainty, agents
maximize Expected utility

$$\begin{array}{l} \text{Max } U(c_1) + \beta E_1[U(c_2)] \\ \text{st. } \left. \begin{array}{l} c_1 + s = y_1 \\ c_2 = y_2 + (1+r)s \end{array} \right\} \end{array}$$

IBC

$$C_1 + \frac{C_2}{1+k_2} = \frac{Y_1}{1+k_2} + \frac{Y_2}{1+k_2}$$

$$C_2 = (1+k_2)Y_1 + Y_2 - (1+k_2)C_1$$

$$\max_{C_1} \log C_1 + \beta E \log [(1+k_2)Y_1 + Y_2 - (1+k_2)C_1]$$

$$\Leftrightarrow \max_{C_1} \log C_1 + \beta \left[\frac{1}{2} (\log [(1+k_2)Y_1 + E(Y_2) + 1] - (1+k_2)C_1) \right. \\ \left. + \frac{1}{2} (\log [(1+k_2)Y_1 + E(Y_2) - 1] - (1+k_2)C_1) \right]$$

FOC

$$\frac{1}{c_1} = \beta(1+r) E_1 \left[\frac{1}{c_2} \right]$$

stochastic Euler equation.

$$c_2 = \begin{cases} \text{high} & \text{prob } 1/2 & c_{2H} \\ \text{low} & \text{--- } 1/2 & c_{2L} \end{cases}$$

— with no risk : $E[c_2]$, $E[y_2]$

Without risk, this is what we
did before $y_2 = E(y_2)$

$$\frac{1}{c_1} = (1+r) \beta \frac{1}{E(c_2)}$$

With risk

$$\frac{1}{c_1} = (1+r) \beta E\left[\frac{1}{c_2}\right]$$

same
 c_1 ?